

VINOD KUMAR CHAGANTI

ASIC Physical Design Engineer

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PROFESSIONAL SUMMARY

Entry-level ASIC Physical Design Engineer with 8 months of intensive hands-on training and project implementation in complete RTL-to-GDSII flow using 28nm technology. Proficient in industry-standard EDA tools including Cadence Genus and Synopsys IC Compiler II (ICC2). Demonstrated expertise across all backend PD stages: Synthesis, Floor-planning, Power Planning, Placement, Clock Tree Synthesis (CTS), Routing, and Physical Verification (DRC/LVS/Antenna). Experienced in Static Timing Analysis (STA) for setup and hold violation closure, ECO implementation, and signal integrity resolution. Strong TCL scripting skills with the ability to automate and optimize Physical Design workflows.

TECHNICAL SKILLS

Physical Design Flow: RTL-to-GDSII / Netlist-to-GDSII, Floor-planning, Power Planning, Placement, CTS, Routing, ECO, Sign-off

Synthesis & Optimization: Logic Synthesis, Area / Power / Timing trade-offs, Constraint (SDC) development

Static Timing Analysis (STA): Setup & Hold analysis, False paths, multi-cycle paths, WNS/TNS closure, DRV fixing, Multi-corner STA

Physical Verification: DRC, LVS, ERC, Antenna rule checking and fixing

EDA Tools: Cadence Genus (Synthesis), Synopsys Design Compiler (DC Shell), Synopsys IC Compiler II (ICC2)

Scripting & OS: TCL (Tool Command Language) scripting for PD automation, Linux / Unix, Windows

KEY PROJECTS

Project 1: Physical Design Implementation – Block 'RP-TOP'

Technology: 28nm | Target Frequency: 454 MHz | Complexity: 70K Standard Cells, 6 Macros | Metal Stack: 9 layers (M1–M9)

- Executed complete RTL-to-GDSII flow: performed logic synthesis using Cadence Genus with SDC constraints, followed by full P&R implementation in Synopsys ICC2.
- Designed a robust floorplan with 9 metal layers (M1–M9); strategically placed 6 macros to minimize wire length, reduce congestion, and meet area targets.
- Performed Clock Tree Synthesis (CTS) to build a balanced, low-skew clock distribution network; met target clock latency and skew budgets across all clock domains.
- Resolved routing congestion hotspots through application of partial blockages, cell density optimization, and selective cell padding; achieved clean routed netlist with zero shorts.
- Achieved multi-corner timing closure (setup and hold) by analyzing WNS/TNS across all PVT corners and applying targeted ECO fixes and buffer insertions.
- Cleared all Physical Verification sign-off checks (DRC, LVS, Antenna) for final GDS tape-out readiness; resolved antenna violations via diode insertion and wire splitting.

Project 2: High-Density Block Implementation – 'TILE'

Technology: 28nm | Target Frequency: 588 MHz | Complexity: 400K+ Standard Cells | 18 Macros

- Led synthesis and full-flow PnR implementation of a high-complexity 400K-instance design using Synopsys DC Shell for synthesis and ICC2 for Place & Route.
- Managed hierarchical floor-planning for 18 macros; designed power grid for IR drop compliance and ensured robust power distribution across all voltage domains.
- Implemented custom SDC path groups and weight preferences to prioritize critical timing paths during placement, improving WNS by focusing optimization effort on critical endpoints.

- Addressed aggressive routing congestion in the 400K+ cell block through strategic density screens, cell padding, and pin access optimization; achieved DRC-clean routing closure.
- Identified and resolved complex post-route DRC violations and antenna effects; used both diode insertion and metal rerouting strategies to achieve full physical verification sign-off.
- Conducted multi-corner STA to validate timing closure across process corners (SS/TT/FF) and temperature/voltage extremes; confirmed zero setup and hold violations at sign-off.

PROFESSIONAL TRAINING:

Advanced ASIC Physical Design Engineer Training (Full-Time) | Sumedha IT, Hyderabad 2024

- Completed an intensive, industry-aligned program covering the complete RTL-to-GDSII backend flow from first principles to sign-off.
- Core curriculum: Logic Synthesis (Genus / DC Shell), Floor-planning & Power Planning, Placement, Clock Tree Synthesis (CTS), Routing, Timing Closure (STA), and Physical Verification (DRC/LVS/Antenna).
- Practical labs: Block-level implementation on Synopsys ICC2 and Cadence Genus with real 28nm technology libraries, SDC constraint writing, and TCL automation scripting.
- Project work: End-to-end implementation of RP-TOP (70K cells) and TILE (400K cells) blocks to tape-out-ready GDS, mimicking industry backend PD workflows.

EDUCATION:

B.Tech – Electronics and Communication Engineering

Sri Mittapalli College of Engineering
CGPA: 7.13 / 10

2020 – 2024

Intermediate (MPC) | Sri Chaitanya Junior College, Telangana

CGPA: 8.15 / 10

2018 – 2020